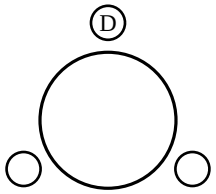
Preview: y2002_q22 (v1)

Counting arrangements: fill places one by one and multiply the choices

Part A: Each question is worth 3 points

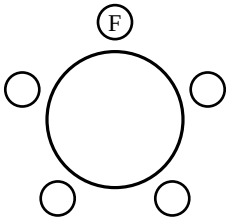
1. Three friends, David, Anna and Grace, go to a restaurant and sit at a table with 3 chairs. David always sits on the same chair (marked D).



In how many different ways can the friends sit around the table?

A. 7 ***B.*** 6 ***C.*** 3 ***D.*** 2 ***E.*** 1

2. Five friends, Felix, John, Grace, Ben and Emma, go to a restaurant and sit at a table with 5 chairs. Felix always sits on the same chair (marked F).



In how many different ways can the friends sit around the table?

A. 24 ***B.*** 25 ***C.*** 30 ***D.*** 120 ***E.*** 121

3. David, Grace and Felix stand in a row for a photo. Each time they stand in a different order. How many different photos can they take?

A. 2 ***B.*** 3 ***C.*** 4 ***D.*** 6 ***E.*** 9

4. Hugo, Emma, Anna and David stand in a row for a photo. Each time they stand in a different order.
How many different photos can they take?

A. 30 ***B.*** 29 ***C.*** 27 ***D.*** 25 ***E.*** 24

Part B: Each question is worth 4 points

5. Four children, Emma, Iris, Hugo and David, sit on a bench. Emma always sits at the left end and Iris always sits at the right end.
In how many different ways can the children sit on the bench?

A. 24 ***B.*** 6 ***C.*** 4 ***D.*** 2 ***E.*** 1

6. Six children, Grace, Clara, Hugo, Emma, Iris and Anna, sit on a bench. Grace always sits at the left end and Clara always sits at the right end.
In how many different ways can the children sit on the bench?

A. 720 ***B.*** 120 ***C.*** 24 ***D.*** 16 ***E.*** 4

7. Grace has 2 T-shirts, 3 pairs of trousers, all different. An outfit is one T-shirt and one pair of trousers.
How many different outfits can Grace make?

A. 5 ***B.*** 6 ***C.*** 7 ***D.*** 9 ***E.*** 12

8. Emma has 4 T-shirts, 3 pairs of trousers and 3 pairs of shoes, all different. An outfit is one T-shirt and one pair of trousers and one pair of shoes.
How many different outfits can Emma make?

A. 10 ***B.*** 12 ***C.*** 13 ***D.*** 15 ***E.*** 36

Part C: Each question is worth 5 points

9. Four children, Anna, Hugo, Grace and Clara, sit in a row of 4 chairs at the cinema. Anna and Hugo are best friends and want to sit next to each other.
In how many different ways can the children sit?

A. 14 ***B.*** 12 ***C.*** 10 ***D.*** 6 ***E.*** 3

10. Five children, Iris, Felix, David, Emma and Anna, sit in a row of 5 chairs at the cinema. Iris and Felix are best friends and want to sit next to each other.
In how many different ways can the children sit?

A. 48 ***B.*** 50 ***C.*** 59 ***D.*** 72 ***E.*** 120

11. Grace paints a flag with three stripes: top, middle and bottom. Grace has 3 colours (red, green and blue). All three stripes must be different colours.
How many different flags can Grace paint?

A. 1 ***B.*** 3 ***C.*** 4 ***D.*** 5 ***E.*** 6

12. Grace paints a flag with three stripes: top, middle and bottom. Grace has 6 colours (black, green, yellow, white, red and orange). All three stripes must be different colours.
How many different flags can Grace paint?

A. 18 ***B.*** 20 ***C.*** 120 ***D.*** 216 ***E.*** 720

Answer Key

1.	**D**	**2.**	**A**	**3.**	**D**	**4.**	**E**	**5.**	**D**	**6.**	**C**
7.	**B**	**8.**	**E**	**9.**	**B**	**10.**	**A**	**11.**	**E**	**12.**	**C**

Solutions

1. D (2) — David's chair is fixed. The other 2 friends fill the other chairs: 2 × 1 = 2 ways. [2002 Q22 · fixed_seat]

2. A (24) — Felix's chair is fixed. The other 4 friends fill the other chairs: 4 × 3 × 2 × 1 = 24 ways. [2002 Q22 · fixed_seat]

3. D (6) — 3 children can stand first, then 2 are left for the second place, and so on: 3 × 2 × 1 = 6. [2002 Q22 · photo_line]

4. E (24) — 4 children can stand first, then 3 are left for the second place, and so on: 4 × 3 × 2 × 1 = 24. [2002 Q22 · photo_line]

5. D (2) — Only the 2 children in the middle can change places: 2 × 1 = 2 ways. [2002 Q22 · ends_fixed]

6. C (24) — Only the 4 children in the middle can change places: 4 × 3 × 2 × 1 = 24 ways. [2002 Q22 · ends_fixed]

7. B (6) — Each T-shirt goes with each pair of trousers: 2 × 3 = 6. [2002 Q22 · outfits]

8. E (36) — Each T-shirt goes with each pair of trousers and each pair of shoes: 4 × 3 × 3 = 36. [2002 Q22 · outfits]

9. B (12) — Think of Anna and Hugo as one block: the block and the other 2 make 3 things to order, 3 × 2 × 1 = 6 ways. Inside the block Anna and Hugo can swap, so 2 × 6 = 12. [2002 Q22 · side_by_side]

10. A (48) — Think of Iris and Felix as one block: the block and the other 3 make 4 things to order, 4 × 3 × 2 × 1 = 24 ways. Inside the block Iris and Felix can swap, so 2 × 24 = 48. [2002 Q22 · side_by_side]

11. E (6) — Top stripe: 3 choices, middle: 2, bottom: 1. 3 × 2 × 1 = 6. [2002 Q22 · stripes]

12. C (120) — Top stripe: 6 choices, middle: 5, bottom: 4. 6 × 5 × 4 = 120. [2002 Q22 · stripes]